TMS component verification

Circuit learning guide and datasheet audit

14 October 2026 | A 2 installed schematic / DRAFT | Engineering draft

120 physical schematic references. 380 pin records, including 59 explicit no-connects. Every part has a purpose, value basis, evidence reference and remaining verification action. The inventory comes from saved native CAD, not the historical BOM. Identical parts share a register row; the CSV lists all 120 references individually.

The installed A 2 schematic corrects documented wiring/support-component defects. The selected-part archive and independent node checks cover the complete schematic. The owner approved a schematic-stage installation exception; the PCB remains empty. Open prototype/layout qualifications prevent unconditional PCB fabrication or live-load approval.

Companion files

COMPONENT_VERIFICATION.md contains the editable audit. component-verification.csv lists every selected order, footprint, purpose, value basis, source, status and remaining check. pin-connections.csv lists all 380 physical pins, including 59 explicit NCs. external-system-parts.csv covers items outside the CAD count.

Value and rating conventions

Exact ordered capacitors/resistors and their manufacturer specifications are now selected. Buck capacitors are 50 V X7R 1210; other selected bulk ceramics are 16 V X5R 0805. Reference electrolytics are 16 V Panasonic FT, with correct 4 mm/5 mm bodies. Precision reference resistors are 0.02%/2 ppm/K; feedback/pressure dividers 0.1%/25 ppm/K; ordinary resistors 1%/100 ppm/K. Typical bias/AC curves support screening calculations but are not guaranteed effective-capacitance limits.

Author-selected assumptions

Owner authorized assumptions: 1%FS calibrated static target near 20-30C calibration window;50 Hz usable band;1000 SPS/channel;1 A 5 V planning load. Pressure output resistance≤100 Ohm and ignition load≤2 A at 8.4 V (cold resistance≥4.2 Ohm) are unmeasured design contracts. They cannot establish flawless performance. Wider temperature accuracy, actual load behavior and all PCB/bench qualifications remain open.

How to interpret the status

The register records author static checks. Each remaining qualification action is still required. Source coverage, clean ERC and matching node groups do not establish physical fit, measured performance or owner acceptance.

Corrections and open qualifications

F01 / Corrected: RF bypass

Affected: C52/U8. 47 pF now implements archived Digi revisionW p52, with 1 uF and 10 uF.

Action: Verify placement and ≤50 mV p-p at the module on the PCB.

F02 / Corrected: unused analog inputs

Affected: U5/U7. Unused ADS1256 AINCOM/AIN2-7 and ADS1220 analog/reference pins 6-10 are explicit NC. CLK3/DGND4/AVSS5 remain GND.

Action: Firmware must select only connected channels; ADS1220 low-side switch/IDAC disabled.

F03 / Selected and calculated: capacitance

Affected: Power and LDO capacitors. Exact KEMET parts/curves are archived. Input 30 uF/output 100 uF nominal banks meet the stated TI equation screen using typical derating models.

Action: Effective capacitance, loop response and temperature still need prototype evidence; simulation curves are not guaranteed limits.

F04 / Corrected: ADC loading

Affected: U5/U10/R46/C44. ADS1256 bufferON replaces bufferOFF. Pressure divider now drives OPA388, then 47 Ohm/10 nF into ADS1220.

Action: BufferON and PGA-bypass configurations are required firmware conditions; verify settling and complete-chain calibration.

F05 / Open qualification: input faults

Affected: D20/D21/D22/D40/D41. The normal contract is correctly wired 0-5 V sensors. TPD2E2U06/BAV199 do not prove sustained battery-miswire or arbitrary off-state protection.

Action: Before powering a prototype with field cables, evaluate actual sequencing/injection and ESD. A 16.8 V bridge-signal fault can exceed the 1k/OPA 10 mA momentary-current screen; it is outside this design contract.

F06 / Selected: reference/ferrite parts

Affected: C32/C34/FB1. Panasonic FT exact capacitance/package/ESR maxima and Murata family specification are archived. C32 footprint corrected to 4x5.8 mm.

Action: OPA350 stability with actual ESR minima, capacitor aging/temperature and ferrite resonance remain PCB/bench qualifications.

F07 / Corrected: switching frequency

Affected: U2. LMR51430YFDDCR FPWM minimum 0.95 MHz exceeds Digi 500 kHz recommendation.

Action: Frequency compliance does not prove RF ripple; measure at U8 during transmit/startup.

F08 / Assumed operating contract: sensors

Affected: BCA-100L/IPSU-GP1K5-6. Retained 5 V bridge excitation; pressure output resistance assumed≤100 Ohm with 133.2k load. Neither assumption adds a manufacturer guarantee.

Action: Calibrate both complete channels.1%FS static target applies near calibration temperature (20-30C proposed), not across the sensor full ambient range.

F09 / Defined: accuracy/bandwidth intent

Affected: Both acquisition channels. Author assumption: 1%FS calibrated static target,50 Hz usable band,1000 SPS per channel. Analog gains at 50 Hz are~0.987 thrust/~0.938 pressure.

Action: Measure sensor+analog+ADC response, independent-clock alignment and dropped samples; static accuracy and AC amplitude are separate.

F10 / Open: physical/current-path qualification

Affected: Connectors/sockets/ignition. Assumed load≤2 A at 8.4 V (cold resistance≥4.2 Ohm); this does not establish actual nichrome current or ignition energy. J5 manufacturer drill mismatch is corrected; J6 is now PCB-mount XT60PW-F.

Action: Resolve purchased AMASS drawing revision, polarity/mating fit, sockets, enclosure, fuse/switch/harness and copper. FullPCB/human review remains open.

F11 / Corrected: unintended EN connection

Affected: J1/D1/U2/R1/R2. Overlapping battery-input and EN wire segments bypassed the enable divider. Separate routed schematic wires now match the intended node groups.

Action: Independent contract compares every physical pin across 69nodes; a deliberate short fault must be rejected.

F12 / Corrected: UART power-off path

Affected: U9/C53/C54/R52-R55. TXU0204 provides specified Ioff/VCC isolation across Teensy3V3 and RF3V3. Idle pulls stop transmit when RF is off.

Action: Check all USB/battery/off/ramp modes; this is common-ground power-off buffering, not galvanic isolation.

F13 / Corrected: supported UART CTS pin

Affected: U1. Radio moved to documented Serial3 TX14/RX15/CTS19/RTS18. Former Serial 2 CTS on D14 was unsupported.

Action: Use Serial3.begin(115200), attachCts(19), attachRts(18). Supported direct CTS avoids XBAR inverted-polarity behavior.

F14 / Corrected: local footprint source lineage

Affected: Five TMS local footprints. All 120 native pin sets match actual footprint pads through the normal shared resolver. Five versioned receipts bind manufacturer evidence, CAD/source hashes, selected orders, derivation notes and every pad function.

Action: Fault tests reject changed CAD/source, wrong part identity, missing pad functions and false owner sign-off. Physical fit and all mandatory human decisions remain open.

F15 / Specified: module silicon revision

Affected: U8. Digi p101 documents an early EFR32MG revisionB brownout/ramp-dip restart failure. RevisionC corrects it; the module label identifies the revision.

Action: Specify/inspect revisionC or later at assembly. The user module silicon revision is unverified; if revisionB must be used, qualify power ramps and complete-powercycle recovery.

F16 / Corrected: connector footprint polarity

Affected: J1/J4/J6. The assigned AMASS footprints mark pad 1 negative and pad 2 positive. A 0/A 1 symbols had these functions reversed; matching pad sets alone did not detect the error.

Action: Corrected schematic positive pin 2/return pin 1, retained component UUIDs, and added a check against actual footprint +/- markings plus the independent electrical node contract. Verify physical molded polarity before assembly.

1. Power: switching, regulation and rail margins

Native sheet: power.kicad_sch

Follow J1 to D1, then U2/L1 and the output capacitor bank. D1 conducts for correct battery polarity. U2 switches at 1.1 MHz; L1 stores energy and the capacitors supply the load between switching/feedback corrections. R3/R4 set the regulated voltage, R5 is a loop-test break point, and C3 supplies the floating gate driver between CB and SW. D2 is a bounded transient clamp after the diode.

At12-16.8 V electronics battery input, protected VBAT is assumed≥11.3 V under the 1 A 5 V load. The 511k/105k enable divider gives ample enable voltage; it is not a4S discharge cutoff. External fusing and battery protection are system requirements.

DC calculation: 4.98175 V nominal; conservative feedback/reference/bias corners 4.89337-5.07041 V. With an assumed≤0.10 V transient deviation and≤12 mV analog distribution drop, +5VA screen is 4.78137-5.17041 V. This fits ADS1256 recommended 4.75-5.25 V, and reference screen≤2.58672 V including 0.02% divider corners and 1 mV U6 offset, below 2.6 V. A waveform measurement must confirm these assumptions.

Input 30 uF nominal screens to 8.16 uF at 16.8 V; output 100 uF screens to 54.43 uF at 5 V using typical bias retention and conservative tolerance/temperature/aging/small-signal factors. TI Eq14 needs 42.11 uF for 0-to 1 A,0.95 MHz and 0.10 V. The larger bank was chosen because the smaller bank failed this screen. Loop response is still unverified.

5 V planning budget: 650 mA Teensy/SD/logic/LEDs,100 mA RF branch,47 mA relay,60 mA analog/bridge/buffers =857 mA against 1 A allocation. These are author reserves, not measured maxima. An 85% efficiency thermal screen gives~0.88 W converter loss at 5 W output; actual two-layer copper/ambient must be assessed. U3 at 100 mA dissipates~0.17 W.

+5V branches through FB1 to analog+5VA, and through U3 to RF+3V3_RF. ADC digital supplies and U9 A-port use Teensy+3V3. Both 3.3 V outputs remain separate. Local bypass placement and the real return paths determine noise performance.

Component register / 1. Power

Values and specifications are read from native CAD. Every reference on power.kicad_sch is listed below.

Reference / value	Purpose and value basis	Evidence / status
C1, C11, C12 (x 3) 10 uF C1210C106K5RACTU	**Buck input switching-current reservoir.** Three 10 uF 50 V X7R 1210 KEMET parts give 30 uF nominal. At16.8 V, screening factors 0.45 bias x 0.90 tolerance x 0.85 temperature x 0.93 aging x 0.85 small-signal factor give 8.16 uF, above TI4.7 uF guidance. Bias/AC curves are typical models, so this is not a guaranteed effective-C limit. **Verify:** Place at VIN/GND; confirm effective C, input RMS current (~0.5 A at 1 A output worst duty) and hot-plug behavior.	KEMET / BUCK **Static check** **Qualify on PCB**
C2 100 nF C0603C104K5RACTU	**High-frequency buck VIN bypass.** 100 nF 50 V X7R follows TI input recommendation alongside the bulk bank. It handles faster switching edges with a short VIN/GND loop. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / BUCK **Static check** **Qualify on PCB**
C3 100 nF C0603C104K5RACTU	**Bootstrap supply between the floating gate-driver pins.** 100 nF 50 V X7R directly connects CB6 to SW 2, as TI section 9.2.2.7 specifies 0.1 uF rated 16 V or higher. It is not returned toGND. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / BUCK **Static check** **Qualify on PCB**
C4, C5, C13, C14, C15, C16, C17, C18, C19, C29 (x 10) 10 uF C1210C106K5RACTU	**Buck output charge reservoir and ripple filtering.** Ten 10 uF 50 V X7R 1210 parts give 100 uF nominal. Same screening model with 0.90 bias retention at 5 V gives 54.43 uF. TI Eq14 needs 42.11 uF for assumed 0-to 1 A step,0.95 MHz and 0.10 V deviation. Predicted capacitive deviation 77 mV is a screen, not proof of loop stability or settling. **Verify:** Verify loop/load-step response, total distributed capacitance and actual PCB parasitics; the bank exceeds TI typical table values deliberately.	KEMET / BUCK **Static check** **Qualify on PCB**
C6 10 uF C0805C106K4PACTU	**Local analog 5 V reservoir after the ferrite.** 10 uF 16 V X5R 0805 exact KEMET order. Provides local load charge; its value is an engineering choice. A ferrite/MLCC network can resonate and must be assessed with the real layout. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / FERRITE / ADC1 / ADC2 **Static check** **Qualify on PCB**
C7, C8 (x 2) 10 uF C0805C106K4PACTU	**LDO input/output bypass and transient reservoir.** 10 uF 16 V X5R 0805 exact order. At5 V a typical 0.45 bias factor with tolerance/temperature/aging/AC screening leaves 2.72 uF; at 3.3 V retention is higher. This clears TLV 755P1 uF input guidance and 0.47 uF effective output minimum in the model. **Verify:** Verify effective capacitance and RF startup/load steps; local total RF capacitance is below the 200 uF performance recommendation.	KEMET / LDO **Static check** **Qualify on PCB**
D1 SS54 SS54	**Blocks reverse battery polarity in normal DC use.** SS54 is a 40 V / 5 A SMA rectifier in the retained MDD sheet. 40 V exceeds 16.8 V DC; transient and thermal margins remain unqualified. P=I*VF. **Verify:** Confirm actual manufacturer; measure input surge, reverse faults and temperature at worst load.	REVERSE **Static check** **Qualify on PCB**
D2 SMBJ18A SMBJ18A-13-F	**Clamps positive input transients after reverse protection.** Unidirectional SMBJ18A-13-F:18 V stand-off exceeds 16.8 V;29.2 V maximum clamp at 20.5 A specified 10/1000us pulse is below LMR36 V operating/38 V abs max and 50 V capacitors. These pulse conditions do not cover every possible cable surge. **Verify:** Cathode 1 toVBAT, anode 2 GND. External fuse mandatory; verify hot-plug waveform and temperature derating.	TVS / BUCK **Static check** **Qualify on PCB**
FB1 BLM21PG601SN1D BLM21PG601SN1D	**Reduces switching-noise coupling into analog 5 V.** BLM21PG601SN1D is 600 Ohm at 100 MHz, not DC. Initial DCRmax 0.14 Ohm gives 8.4 mV at 60 mA;0.20 Ohm post-test bound gives 12 mV. Current rating has ample margin for 60 mA. No ADC noise attenuation is inferred from the 100 MHz rating. **Verify:** Verify actual analog supply noise/resonance and PCB return paths.	FERRITE / ADC1 / ADC2 **Static check** **Qualify on PCB**

Reference / value	Purpose and value basis	Evidence / status
J1 XT30PW-F XT30PW-F	**Electronics 4S / separateignition 2S battery connectors.** XT30PW-F selected drawings/5mmpowerpitch. Input J1return isGND;J4return is IGN_GND. Pad 2 is positive and pad 1 is return, matching the assigned KiCad footprint +/- markings. Connector family 15 A rating exceeds planning currents; mating gender/polarity and inconsistent retention-hole drawings require exact purchased-part fit verification. **Verify:** Do not use genericcatalog dimensions interchangeably with V 1.2; verify harness/polarity and external fuse.	XT30 **Static check** **Qualify on PCB**
L1 3.3 uH / SRP1038A-3R3M SRP1038A-3R3M	**Stores switching energy between SW and 5 V.** SRP1038A-3R3M:3.3uH +/-20%,11.8 mOhm maxDCR,Irms 11 A/Isat 20 A at 25C. TI1.1 MHz/5 V table uses 3.3uH. At16.8 V, Lmin 2.64uH/fmin 0.95 MHz gives 1.40 A p-p and 1.70 A peak at 1 A output, below IC min peak limit 3.67 A. **Verify:** Check hot inductance/loss and switching layout; this physically large inductor favors margin over compactness.	IND / BUCK **Static check** **Qualify on PCB**
R1 511 kOhm CRCW0603511KFKEA	**Upper enable divider.** 511k with 105k gives EN about 1.93 V at assumed 11.3 V protected input, above 1.36 V max rising threshold. The 1% limits and 200nA EN source current keep startup safely below normal 12 V battery operation. Divider is not a4S battery cutoff. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	CRCW / BUCK **Static check** **Qualify on PCB**
R2 105 kOhm CRCW0603105KFKEA	**Lower enable divider.** 105k completes the 511k/105k divider. Nominal rising threshold 1.227*(1+511/105)=7.20 V before EN source current; normal input 12-16.8 V is comfortably above this. External battery undervoltage protection is required. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	CRCW / BUCK **Static check** **Qualify on PCB**
R3 100 kOhm TNPW0603100KBEEA	**Sets the main 5 V feedback ratio.** R3=100k and R4=13.7k, both 0.1%/25 ppm/K. Include R5=49.9 Ohm: VOUT=0.600*(1+100049.9/13700)=4.98175 V. TI0.591-0.609 V reference, resistor tolerances and conservative +/-50nA FB bias give 4.89337-5.07041 V DC. **Verify:** Verify DC voltage at the load and ≤0.10 V transient screen; include distribution drop before accepting ADS1256 supply/reference windows.	TNPW / BUCK **Static check** **Qualify on PCB**
R4 13.7 kOhm TNPW060313K7BEEA	**Sets the main 5 V feedback ratio.** R3=100k and R4=13.7k, both 0.1%/25 ppm/K. Include R5=49.9 Ohm: VOUT=0.600*(1+100049.9/13700)=4.98175 V. TI0.591-0.609 V reference, resistor tolerances and conservative +/-50nA FB bias give 4.89337-5.07041 V DC. **Verify:** Verify DC voltage at the load and ≤0.10 V transient screen; include distribution drop before accepting ADS1256 supply/reference windows.	TNPW / BUCK **Static check** **Qualify on PCB**
R5 49.9 Ohm CRCW060349R9FKEA	**Control-loop injection break point in feedback sensing.** 49.9 Ohm is retained as a small series loop-test resistor. Its DC contribution to the upper feedback leg is included. C9 feed-forward is removed because the new regulator example does not require the old 75 pF network. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	CRCW / BUCK **Static check** **Qualify on PCB**
TP1 VBAT VBAT	**Probe protected battery and regulated 5 V.** 2 mm pad is a probing/assembly choice, not an electrical datasheet requirement. TP1 should show protected VBAT; TP2 about 4.981 V unloaded nominal. **Verify:** Check probe access/clearance; measure load and startup waveforms without shorting adjacent nodes.	BUCK **Probe geometry**
TP2 +5V +5V	**Probe protected battery and regulated 5 V.** 2 mm pad is a probing/assembly choice, not an electrical datasheet requirement. TP1 should show protected VBAT; TP2 about 4.981 V unloaded nominal. **Verify:** Check probe access/clearance; measure load and startup waveforms without shorting adjacent nodes.	BUCK **Probe geometry**
U2 LMR51430YFDDCR LMR51430YFDDCR	**Converts protected 4S power to the main 5 V rail.** LMR51430YFDDCR is the 1.1 MHz FPWM DDC6 order. Frequency minimum 0.95 MHz exceeds Digi 500 kHz recommendation; 4.5-36 V operation covers protected 11.3-16.8 V. Board planning limit remains 1 A. Pin 1GND/2SW/3 VIN/4FB/5EN/6CB. **Verify:** Verify actual 5 V/load steps and temperature; no 3 A board rating follows from this IC.	BUCK / RF **Static check** **Qualify on PCB**

Reference / value	Purpose and value basis	Evidence / status
U3 TLV75533PDBVR TLV75533PDBVR	**Provides an independent 3.3 V supply for the XBee.** TLV75533PDBVR is the fixed 3.3 V DBV package. At 100 mA, (4.981-3.3)*0.1 is about 0.168 W. The 500 mA IC rating is not this board's qualified output. **Verify:** Check DBV thermal resistance/copper, dropout, inrush, output discharge and RF ripple.	LDO / RF **Static check** **Qualify on PCB**

2. Controller: timing, logging and supported pins

Native sheet: controller.kicad_sch

Teensy 4.1 owns its MCU support circuits, clock, regulator and microSD socket. The board needs removable socket rows and local VIN reservoir, not duplicate bare-MCU circuitry. Module pad names are an explicit local convention mapped to the PJRC card.

Thrust uses SPI0 D11MOSI/D12MISO/D13SCLK, CS10 and DRDY9. Pressure uses SPI1 D26MOSI/D1MISO/D27SCLK, CS0 and DRDY2. Independent DRDY timestamps are required because the ADC clocks differ. The 1kSPS target is not a claim of zero dropped samples.

Radio uses Serial3: TX14/RX15/CTS19 and RTS18. PJRC explicitly supports pin 19 as Serial3 CTS and any digital pin for RTS. Use Serial3.begin(115200), then attachCts(19) and attachRts(18). CTS high pauses transmit. UART payload at 60frames/s x 32bytes x 10bits is 19.2kbit/s before protocol overhead, below 115.2kbit/s; RF airtime/retries and SD stalls still require testing.

JP1 connects battery 5 V to VIN. For simultaneous USB/battery, remove JP1 or cut the Teensy VIN-VUSB link per PJRC. U9 prevents direct UART drive into an unpowered RF module. USB alone leaves the analog ADC rails off, so acquisition must wait for valid analog power.

D3 controls the relay driver and must start low. D4/D5/D6 drive 1k LED resistors. Approximate currents 1.4/1.55/1.45 mA use typical LED forward voltage; real brightness and GPIO VOH vary. The red LED means commanded state. There is no physical ARM feedback.

Component register / 2. Controller

Values and specifications are read from native CAD. Every reference on controller.kicad_sch is listed below.

Reference / value	Purpose and value basis	Evidence / status
C10 10 uF C0805C106K4PACTU	**Local Teensy VIN reservoir.** 10 uF 16 V X5R 0805 is an added board-level reservoir near the module VIN. Module-owned regulation/decoupling remains on Teensy; the exact 10 uF value is an engineering transient choice. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / MCU **Static check** **Qualify on PCB**
D10 GREEN APTD1608LCGCK	**Green status indication.** APTD1608LCGCK is a low-current 0603 green LED characterized at 2 mA. Selected current is lower, so datasheet brightness at 2 mA is not guaranteed here. **Verify:** Confirm polarity, visibility in sunlight, thermal and mechanical assembly.	GREEN **Static check** **Qualify on PCB**
D11 RED APTD1608LSURCK	**Red command indication.** APTD1608LSURCK low-current red matches the approximately 1.55 mA design. No physical ARM feedback exists in this circuit. **Verify:** Confirm polarity and firmware meaning; mark user interface accordingly.	RED **Static check** **Qualify on PCB**
D12 YELLOW APTD1608LSYCK	**Yellow telemetry activity indication.** APTD1608LSYCK low-current yellow matches approximately 1.45 mA design. Duty cycle is firmware-controlled. **Verify:** Confirm polarity and visible duty cycle.	YELLOW **Static check** **Qualify on PCB**
JP1 VIN POWER LINK TSW-102-07-L-S	**Removable battery feed to Teensy VIN.** Samtec TSW-102-07-L-S is a real 2-pin 2.54 mm header. A separate conductive shunt is an assembly item. Remove shunt for simultaneous USB/battery unless Teensy VIN-VUSB link is cut. This is a configuration choice, not automatic power arbitration. **Verify:** Verify mating shunt rating and the chosen USB configuration.	SHUNT_HEADER / MCU **Static check** **Qualify on PCB**
R10 1 kOhm CRCW06031K00FKEA	**Limits green LED/GPIO current.** 1 kOhm gives about (3.3-1.9)/1000=1.4 mA using the LED's typical VF at 2 mA; actual current uses GPIO VOH and LED curve. **Verify:** Check visibility and worst GPIO current, VF and resistor tolerance; approximate dissipation is 2 mW.	CRCW / GREEN / MCU **Static check** **Qualify on PCB**
R11 1 kOhm CRCW06031K00FKEA	**Limits red LED/GPIO current.** 1 kOhm gives about (3.3-1.75)/1000=1.55 mA with typical VF. Red indicates commanded state, not measured ARM or contact state. **Verify:** Verify brightness and GPIO limits; do not treat the LED as a physical inhibit indication.	CRCW / RED / MCU **Static check** **Qualify on PCB**
R12 1 kOhm CRCW06031K00FKEA	**Limits yellow LED/GPIO current.** 1 kOhm gives about (3.3-1.85)/1000=1.45 mA with typical VF. Low-current LED is chosen for visibility without large GPIO load. **Verify:** Check actual current and behavior while transmitting; select exact resistor.	CRCW / YELLOW / MCU **Static check** **Qualify on PCB**
TP3 +3V3 +3V3	**Probe Teensy-derived ADC logic supply.** 2 mm pad exposes +3V3. This rail is separate from +3V3_RF; its ADC/pull-up allocation is less than 10 mA, to be measured. **Verify:** Verify no two regulator outputs are joined and analog/digital sequencing remains within input limits.	MCU **Probe geometry**
U1 TEENSY41 TEENSY41	**Acquisition, SD logging, telemetry and command processor.** Owner-selected Teensy 4.1. SPI0 D11/D12/D13, CS10/DRDY9; SPI1 D26/D1/D27, CS0/DRDY2. Radio uses documented Serial3 TX14/RX15/CTS19, RTS18. Serial 2 CTS on formerD14 was unsupported and is corrected. All I/O3.3 V. **Verify:** Use Serial3.begin(115200), attachCts(19), attachRts(18); verify buffers, timing, SD peaks and reset defaults. Official UART page is cited separately.	MCU **Static check** **Qualify on PCB**

3. Thrust: bridge, filtering and buffered conversion

Native sheet: thrust.kicad_sch

BCA-100L is the confirmed 100 kgf,2 mV/V +/-10% sensor. At~5 V excitation, nominal full-load signal is~10 mV. CAS recommends 10 V; retained 5 V halves signal amplitude, so datasheet noise/accuracy cannot simply be assumed unchanged. Input/output resistances are 400 Ohm +/-20 Ohm and 350 Ohm +/-3.5 Ohm. Two-point calibration is required.

J2 carries excitation, signal pair and shield. R20/R21 and C20 form a differential low-pass: 1/[2pi(1k+1k+350)220 nF] ~=308 Hz. OPA2388 unity followers avoid loading the bridge. Offset max 7.5uV per amplifier gives 15uV worst differential (~0.15% of 10 mVFS) before calibration; input/clamp leakage and temperature must also be budgeted.

Each OPA output drives a47 Ohm series resistor and 10 nF differential local reservoir at U5. Required firmware setting is ADS1256 bufferON. At1000 SPS, typical 20MOhm differential input impedance gives~4.7 ppm resistive loading from 94 Ohm; the former bufferOFF/PGA 64 typical 4.7 kOhm model gave~1.96% loss. Switched-capacitor settling still needs validation.

Use AIN0-AIN1, PGA 64,1000 SPS and 7.68 MHz. Differential full scale is +/-2 VREF/64 ~=+/-77.8 mV, so the bridge fits with ample overload headroom. Buffer common mode must remain between 0 V and AVDD-2 V; normal bridge common mode nearVEXC/2 fits the screened supply window.

Y1 exact order is ABLS-7.680MHZ-D-4-Y-T with 18 pF load. Two 30 pF capacitors give 15 pF in series; assumed 3 pF parasitic yields 18 pF. Verify frequency, cold/hot startup and drive on the real PCB.

The protection network is coarse cable ESD/rail steering. BAV199 VF can exceed an IC rail-overdrive limit. Normal correctly wired sensor operation is the contract; sustained battery-to-signal faults and off-state injection remain explicit prototype checks.

Why sample rate is not bandwidth

Frequency	Thrust RC gain (308 Hz)	Pressure RC gain (136 Hz)
10 Hz	0.999	0.997
50 Hz	0.987	0.938
100 Hz	0.951	0.805
250 Hz	0.776	0.478
500 Hz	0.524	0.262

First-order analog estimates only. ADC digital filters, sensor dynamics, tolerances and input loading are excluded; these numbers are not the full system transfer function.

Component register / 3. Thrust

Values and specifications are read from native CAD. Every reference on thrust.kicad_sch is listed below.

Reference / value	Purpose and value basis	Evidence / status
C20 220 nF C0603C224K4RACTU	**Differential bridge low-pass capacitor.** 220 nF 16 V X7R 0603 exact KEMET order. With 1k+1k+350 Ohm sensor source, nominal pole 307.84 Hz. Differential DC is near 10 mV, so high common-mode voltage is not applied across this capacitor. Dielectric/tolerance/AC variation still affect the pole. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / BUF / ADC1 **Static check** **Qualify on PCB**
C21 100 nF C0603C104K5RACTU	**Local dual-buffer supply bypass.** 100 nF matches OPAx388 power-supply guidance and serves the shared supply pins of both channels. It must be near pin 8/pin 4 physically. **Verify:** Verify effective capacitance and short return path; additional bulk depends on distribution impedance.	KEMET / BUF **Static check** **Qualify on PCB**
C22 10 nF VJ0603A103JXACW1BC	**Local differential reservoir across thrust ADC inputs.** 10 nF 50 V C0G 5%0603 actual Vishay order. With 47+47 Ohm, nominal pole 169 kHz; upstream 220 nF network sets the intentional low-frequency analog pole. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	VJ / ADC1 **Static check** **Qualify on PCB**
C23 100 nF C0603C104K5RACTU	**ADS1256 analog high-frequency supply bypass.** 100 nF ceramic at AVDD1 follows TI Fig. 25 / bypass guidance, with C27 bulk in parallel. **Verify:** Place at pin 1-to-AGND2 with low loop inductance.	KEMET / ADC1 **Static check** **Qualify on PCB**
C24 100 nF C0603C104K5RACTU	**ADS1256 digital high-frequency bypass.** 100 nF ceramic at DVDD16 follows TI Fig.25 / bypass guidance. RESET15 is not a supply pin. **Verify:** Place at pin 16-to-DGND17; preserve the pin mapping independently of ERC.	KEMET / ADC1 **Static check** **Qualify on PCB**
C25, C26 (x 2) 30 pF C0603C300J5GACTU	**Crystal load capacitors.** Two 30 pF C0G capacitors yield C1*C2/(C1+C2)+Cstray=15 pF+3 pF=18 pF. 3 pF total parasitic is an assumption, not a measured fact. **Verify:** Measure frequency/startup with a low-capacitance technique; adjust only after evaluating real parasitics and drive.	KEMET / XTAL / ADC1 **Static check** **Qualify on PCB**
C27 10 uF C0805C106K4PACTU	**Thrust ADC analog bulk bypass.** 10 uF 16 V X5R 0805 complements the 100 nF supply bypass in TI basic connection circuit. Exact bias behavior is archived; nominal 10 uF is not claimed to remain 10 uF in operation. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / ADC1 **Static check** **Qualify on PCB**
C28 1 uF C0603C105K4RACTU	**Thrust ADC digital reservoir.** 1 uF 16 V X7R 0603 is an engineering addition to 100 nF DVDD bypass. It does not replace C24 at DVDD16. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / ADC1 **Static check** **Qualify on PCB**
D20 TPD2E2U06DCKR TPD2E2U06DCKR	**Coarse cable ESD suppression on the bridge pair.** TPD2E2U06DCKR is the actual 3-pin SC70 part: IO1=1, IO2=2, GND=3. Its low capacitance/leakage suit signals, but transient clamp voltage is above precision-ADC rail limits. **Verify:** Coordinate with series R and rail clamps; test cable ESD. Not sustained miswire or precise input overvoltage protection.	ESD **Static check** **Qualify on PCB**
D21, D22 (x 2) BAV199 BAV199	**Low-leakage steering diodes to analog rails.** BAV199 steering pair: pin 1 anode toGND,pin 2 cathode to+5VA,pin 3 midpoint tosignal. Maxreverseleakage 5nA at 25C/80nA at 150C, VR75 V; maxVF0.9 V at 1 mA exceeds ADC rail-overdrive limits. It is not a guaranteed sustained-fault clamp. **Verify:** Normal 0-5 V sensor harness only. Qualify cable ESD, powered/off-state injection and sustainedmiswire separately; do not infer absolute-max protection from current alone.	CLAMP / BUF / ADC1 **Static check** **Qualify on PCB**

Reference / value	Purpose and value basis	Evidence / status
J2 B5B-XH-A B5B-XH-A	**Load-cell excitation, signal pair and shield connector.** B5B-XH-A provides five defined harness contacts. CAS may supply four or six conductors plus shield; external remote-sense termination must be defined. **Verify:** Verify wire colours against the actual unit, remote-sense handling, shield/earth policy and 1:1 fit.	JST / BRIDGE **Static check** **Qualify on PCB**
R20, R21 (x 2) 1 kOhm CRCW06031K00FKEA	**Limit fault current and form differential input low-pass filter.** Each 1 kOhm plus 220 nF differential C20 and about 350 Ohm bridge output gives fc=1/[2*pi*(2000+350)*220n]=307.8 Hz. Values are engineering choices, not CAS/TI required values. **Verify:** 1k normal-signal filter choice. A 16.8 V-to-signal miswire can exceed OPA 10 mA momentary input limit; excluded from verified operating contract and remains an open fault-qualification finding.	CRCW / BRIDGE / BUF / ADC1 **Static check** **Qualify on PCB**
R22, R23 (x 2) 47 Ohm CRCW060347R0FKEA	**Isolates buffer outputs from differential ADC charge pulses.** 47 Ohm per leg follows TI-style local ADC drive damping. With 10 nF differential C22, the nominal additional pole is 169 kHz. ADS1256 bufferON avoids the old bufferOFF4.7 kOhm typical loading (~1.96% attenuation). **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	CRCW / ADC1 / BUF **Static check** **Qualify on PCB**
R24, R25, R26 (x 3) 33 Ohm CRCW060333R0FKEA	**Damp SPI edge ringing on SCLK, DIN and DOUT.** 33 Ohm is a provisional series damping value. TI ADS1256 discusses small series resistors around 100 Ohm; 33 Ohm must be justified with actual driver/trace impedance and timing. **Verify:** Check source-end placement for each direction and scope edges at ≤1.92 MHz. These resistors are not 5 V fault protection.	CRCW / ADC1 / MCU **Static check** **Qualify on PCB**
R27 10 kOhm CRCW060310K0FKEA	**Keeps thrust ADC deselected while MCU is high impedance.** 10 kOhm pulls CS to DVDD. ADS1256 input leakage can be up to 10 uA; 0.1 V drop leaves about 3.2 V, above 0.8*3.3=2.64 V VIH in this simple static check. **Verify:** Confirm all rail/firmware states and MCU-low sink current (~0.33 mA).	CRCW / ADC1 **Static check** **Qualify on PCB**
TP4 LC_DRDY_N LC_DRDY_N	**Probe thrust DRDY timing.** 2 mm pad on DRDY allows conversion-period and timestamp checks. Nominal 1000 SPS means ~1 ms periods, not guaranteed sample alignment with the other ADC. **Verify:** Check probe loading and firmware missed-sample accounting.	ADC1 **Probe geometry**
U4 OPA2388IDR OPA2388IDR	**Two unity buffers separate bridge impedance from ADC switched input.** OPA2388 works on +5VA and accepts approximately mid-supply common-mode signals. With output-to-inverting-input feedback each channel has nominal gain 1. Offset mismatch remains differential error. **Verify:** Check +5VA 2.5-5.5 V range, noise/offset drift, output settling and capacitive-load stability with R22/R23/C22.	BUF / ADC1 **Static check** **Qualify on PCB**
U5 ADS1256IDBR ADS1256IDBR	**Differential thrust conversion.** ADS1256IDBR: AVDD1/DVDD16/RESET15. Use AIN0-AIN1, gain 64, bufferON,1000 SPS and 7.68 MHz. BufferON typical input impedance 20MOhm at 1000 SPS reduces 94 Ohm resistive loading to 4.7 ppm; this estimate is not a guaranteed switched-capacitor model. Unused analog pins 5/8-13 are NC. **Verify:** Verify input common-mode below AVDD-2 V, calibration, noise and timing. ADC bufferON is a required firmware condition.	ADC1 **Static check** **Qualify on PCB**
Y1 7.680 MHz / CL 18 pF ABLS-7.680MHZ-D-4-Y-T	**Sets ADS1256 conversion and serial timing.** ABLS-7.680MHZ-D-4-Y-T uses standard 18 pF load. 7.68 MHz matches TI data-rate tables. Family sheet alone is not proof of received exact order. **Verify:** Confirm exact order, drive level, ESR, cold/hot startup and frequency. Datasheet rated drive typical 100 uW/max 1000 uW.	XTAL / ADC1 **Static check** **Qualify on PCB**

4. Reference: why ratiometry works

Native sheet: reference.kicad_sch

R30/R31 divide the actual analog excitation supply by two. Since both bridge output and ADC reference scale with excitation, slow excitation changes largely cancel in the digitized ratio. This cancellation depends on matched time response and return paths; it does not remove all switching noise.

Both 10k parts are actual TNPU0.02%/2 ppm/K orders. Independent tolerance corners give ratios 0.4999-0.5001. They are separate resistors, not a specified matched network. Initial ratio and amplifier offset are calibrated; temperature/aging remain in the budget.

R32=10k and C30=1 uF filter the divider before U6. Divider source resistance 5k adds to R32, giving nominal 15ms. Start with≥150ms after rails settle, then run ADC calibration and verify actual settling. Capacitance tolerance/bias means nominal time constant is not a guarantee.

OPA350 buffers the divider for ADS1256 reference current. A bare divider cannot be treated as a low-impedance precision source. The 147 uF aluminum bank plus 100 nF follows TI Fig 26. Panasonic FT exact ESR maxima/package sizes are now known; minimum ESR and closed-loop stability across real parts/temperature still require qualification. Place the bank at U5 reference pins.

The conservative high-rail/ratio/offset screen givesVREF≤2.58672 V against the ADS1256 limit 2.6 V. This depends on the≤0.10 V supply-transient assumption. Validate it with the complete load and layout.

Component register / 4. Reference

Values and specifications are read from native CAD. Every reference on reference.kicad_sch is listed below.

Reference / value	Purpose and value basis	Evidence / status
C30 1 uF C0603C105K4RACTU	**Reference-divider low-pass reservoir.** 1 uF 16 V X7R 0603 exact order. With 10k R32 and 5k divider source, nominal tau 15ms; initial ≥150ms wait is a screening choice, then wait for stable rails/calibration. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / ADC1 / REF **Static check** **Qualify on PCB**
C31 100 nF C0603C104K5RACTU	**Local reference-buffer supply bypass.** 100 nF is used in ADS1256 Fig.26 and OPA350 layout guidance. OPA350 earlier overview mentions 10 nF; this application follows the specific reference circuit and 0.1 uF layout guidance. **Verify:** Place at U6 pin 7-to-pin 4 with a short ground return.	KEMET / ADC1 / REF **Static check** **Qualify on PCB**
C32 47 uF / 16 V AL EEEFT1C470AR	**Low-frequency reference output reservoir.** Panasonic EEEFT1C470AR:47 uF 16 V,+/-20%,4x5.8 mm. Corrects the old provisional 5 mm footprint. MaxESR0.85 Ohm at 100 kHz 20C, ripple 160 mA;105C/2000h rating. TI Fig 26 uses 47 uF aluminum here. **Verify:** Actual minimum ESR, layout and op-amp loop response remain prototype checks.	FT / ADC1 / REF **Static check** **Qualify on PCB**
C33 100 nF C0603C104K5RACTU	**High-frequency reference output bypass.** 100 nF ceramic follows TI Fig.26, handling faster reference charge pulses alongside the electrolytics. **Verify:** Place near VREFP/VREFN; check return path and bank stability.	KEMET / ADC1 / REF **Static check** **Qualify on PCB**
C34 100 uF / 16 V AL EEEFT1C101AR	**Reference output reservoir alongside 47 uF/100 nF.** Panasonic EEEFT1C101AR:100 uF 16 V,+/-20%,5x5.8 mm. MaxESR0.36 Ohm at 100 kHz 20C,ripple 240 mA;105C/2000h. Follows TI Fig 26 aluminum-capacitor topology. **Verify:** Check reference settling and stability across temperature; no ceramic substitution is authorized by matching capacitance alone.	FT / ADC1 / REF **Static check** **Qualify on PCB**
R30, R31 (x 2) 10 kOhm TNPU060310K0HWEA00	**Precision half-excitation reference divider.** Both 10k TNPU060310K0HWEA00:0.02%,2 ppm/K. Nominal ratio 0.5. Independent tolerance corners give 0.4999-0.5001; worst ratio drift 2 ppm/K (not guaranteed matching). Dissipation~0.62mW each is small versus 0.1 W 70C rating. **Verify:** Two-point calibration absorbs initial scale; budget temperature, aging and U6 offset separately.	TNPU / ADC1 / REF **Static check** **Qualify on PCB**
R32 10 kOhm CRCW060310K0FKEA	**Filters divider noise and isolates reference-buffer input.** 10 kOhm copies the series reference-filter resistor in TI Fig.26. With divider source 5k and C30=1 uF, tau=15 ms. **Verify:** Include U6 input bias/leakage error and validate settling; do not use an ideal zero-impedance source model.	CRCW / ADC1 / REF **Static check** **Qualify on PCB**
TP5 LC_VREF LC_VREF	**Probe the ADC reference source.** 2 mm pad exposes nominalVEXC/2 (~2.49 V). A low-bandwidth DMM check does not establish dynamic reference cleanliness. **Verify:** Measure DC ratio, settling and ripple using high-impedance/low-capacitance probes.	ADC1 / REF **Probe geometry**
U6 OPA350UA OPA350UA	**Buffers divided excitation for the ADC reference input.** OPA350UA follows the ADS1256 recommended reference-buffer topology. A bare divider would load into typical 18.5 kOhm reference impedance; output-to-inverting-input feedback gives nominal unity gain. **Verify:** Check offset, input bias, +5VA supply, temperature grade, output-bank ESR/stability and startup.	REF / ADC1 **Static check** **Qualify on PCB**

5. Pressure: scaling and low-impedance ADC drive

Native sheet: pressure.kicad_sch

The confirmed IPSU-GP1K5-6 is 0-1500 psi,0-5 V and 9-32 V supply. It runs from protected 4S VBAT, not 5 V. Board J3 is supply/signal/return; sensor DIN is supply/return/signal, so the harness deliberately maps 1/3/2. The sensor 32 V supply rating does not enlarge the board input rating.

100k/33.2k0.1% divider gives 0.249249 ratio and 1.246246 V for 5 V. It draws 37.54uA from the sensor. Output resistance is unspecified; assumed≤100 Ohm adds≤0.075% scale loss. C40=47 nF C0G 1206 with 24.925k source gives 135.86 Hz nominal pole.

OPA388 U10 buffers this filtered high-impedance node. Max 400pA input bias over 0-85C gives<10uV source error. The output drives 47 Ohm/10 nF at U7, providing ADC charge locally. Without U10, the ADS1220 datasheet provides only typical analog input-current behavior, so a guaranteed high-source-resistance loading calculation was unavailable.

ADS1220 gain 1/PGA bypass allows AIN0-to-AVSS operation. Use internal 2.048 V reference and normal 1000 SPS; IDAC/low-side switch OFF. The signal uses 60.85% of positive ADC range. Separate 100 nF capacitors bypass AVDD12 and DVDD13. CLK3/DGND4/AVSS5 are grounded; unused 6-10 are NC.

Pressure sensor nonlinearity/hysteresis is<0.25%FS BFSL, setting errors<0.5%FS, thermal zero shift<0.04%FS/C and span shift<0.015%/C. The internal ADC reference maxdrift 30 ppm/C also matters. A 1%FS static target is plausible near two-point calibration, proposed 20-30C; it is not guaranteed across the sensor entire ambient range. Include temperature compensation or narrow the qualified temperature window based on tests.

Why sample rate is not bandwidth

Frequency	Thrust RC gain (308 Hz)	Pressure RC gain (136 Hz)
10 Hz	0.999	0.997
50 Hz	0.987	0.938
100 Hz	0.951	0.805
250 Hz	0.776	0.478
500 Hz	0.524	0.262

First-order analog estimates only. ADC digital filters, sensor dynamics, tolerances and input loading are excluded; these numbers are not the full system transfer function.

Component register / 5. Pressure

Values and specifications are read from native CAD. Every reference on pressure.kicad_sch is listed below.

Reference / value	Purpose and value basis	Evidence / status
C40 47 nF C1206C473J5GACTU	**Low-frequency pressure filter at buffered divider.** 47 nF 50 V C0G 5% KEMET C1206C473J5GACTU in 1206. The old 0603 package specification lacked a supported exact order and is explicitly changed. With 24.925k Thevenin source, pole 135.86 Hz. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / ADC2 **Static check** **Qualify on PCB**
C41 100 nF C0603C104K5RACTU	**ADS1220 analog supply bypass.** 100 nF follows section 9.3.3 and is independent of its digital bypass. **Verify:** Place close to AVDD12-to-AVSS5; verify local supply noise.	KEMET / ADC2 **Static check** **Qualify on PCB**
C42 100 nF C0603C104K5RACTU	**ADS1220 digital supply bypass.** 100 nF follows section 9.3.3, separate from AVDD bypass. **Verify:** Place close to DVDD13-to-DGND4; verify logic rail sequencing/input limits.	KEMET / ADC2 **Static check** **Qualify on PCB**
C43 100 nF C0603C104K5RACTU	**Pressure buffer supply bypass.** 100 nF 50 V X7R 0603 at OPA388 V+5-to-V-2 follows TI layout/supply bypass guidance. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / BUF **Static check** **Qualify on PCB**
C44 10 nF VJ0603A103JXACW1BC	**Supplies local pressure ADC charge pulses.** 10 nF 50 V C0G 5%0603 across AIN0/GND; driven through 47 Ohm by U10. Keeps the switched-capacitor load local instead of on 24.925k divider. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	VJ / ADC2 **Static check** **Qualify on PCB**
D40 BAV199 BAV199	**Pressure input rail steering clamp.** BAV199 steering pair: pin 1 anode toGND,pin 2 cathode to+5VA,pin 3 midpoint tosignal. Maxreverseleakage 5nA at 25C/80nA at 150C, VR75 V; maxVF0.9 V at 1 mA exceeds ADC rail-overdrive limits. It is not a guaranteed sustained-fault clamp. **Verify:** Normal 0-5 V sensor harness only. Qualify cable ESD, powered/off-state injection and sustainedmiswire separately; do not infer absolute-max protection from current alone.	CLAMP / ADC2 **Static check** **Qualify on PCB**
D41 TPD2E2U06DCKR TPD2E2U06DCKR	**Coarse pressure-cable ESD suppression.** One channel of TPD2E2U06DCKR protects P_RAW; the unused protection channel is explicitly NC. Protection is not a sustained battery-short rating. **Verify:** Verify DCK pin 3 ground and cable fault/ESD coordination with R40/D40.	ESD **Static check** **Qualify on PCB**
J3 B3B-XH-A B3B-XH-A	**Pressure sensor power, signal and return harness.** B3B-XH-A provides three contacts. PCB order 1 supply/2 signal/3 return deliberately differs from sensor DIN order 1 supply/2 return/3 signal. **Verify:** Verify actual mating cable and sensor pinout before power. Sensor 9-32 V rating does not expand board 4S limits.	JST / SENSOR **Static check** **Qualify on PCB**
R40 100 kOhm TNPW0603100KBEEA	**Scales 0-5 V pressure to ADC range.** 100k/33.2k0.1%25 ppm/K TNPW gives k0.249249;5 V becomes 1.246246 V (60.85% of 2.048 V range). Nominal sensor load 133.2k,37.54uA at 5 V. Opposing tolerances yield +/-0.1502% ratio error; assumed 100 Ohm sensor output adds 0.075% gain loss. **Verify:** Sensor output impedance is not specified; full-chain calibration and loaded-output verification remain required.	TNPW / ADC2 / SENSOR **Static check** **Qualify on PCB**
R41 33.2 kOhm TNPW060333K2BEEA	**Scales 0-5 V pressure to ADC range.** 100k/33.2k0.1%25 ppm/K TNPW gives k0.249249;5 V becomes 1.246246 V (60.85% of 2.048 V range). Nominal sensor load 133.2k,37.54uA at 5 V. Opposing tolerances yield +/-0.1502% ratio error; assumed 100 Ohm sensor output adds 0.075% gain loss. **Verify:** Sensor output impedance is not specified; full-chain calibration and loaded-output verification remain required.	TNPW / ADC2 / SENSOR **Static check** **Qualify on PCB**

Reference / value	Purpose and value basis	Evidence / status
R42, R43, R44 (x 3) 33 Ohm CRCW060333R0FKEA	**Damp pressure SPI edges.** 33 Ohm is a board signal-integrity starting value; no ADS1220 specification requires it. Must not consume SCLK setup/hold margins. **Verify:** Scope at actual bus speed; place each series resistor near its driving endpoint.	CRCW / ADC2 / MCU **Static check** **Qualify on PCB**
R45 10 kOhm CRCW060310K0FKEA	**Keeps pressure ADC deselected during MCU startup.** 10 kOhm pulls CS to the same DVDD logic rail. Static low-state current is about 0.33 mA; high-state margin depends on maximum leakage. **Verify:** Check input leakage/VIH over temperature and all power/reset states.	CRCW / ADC2 **Static check** **Qualify on PCB**
R46 47 Ohm CRCW060347R0FKEA	**Isolates pressure buffer from the ADC reservoir.** 47 Ohm 1%0603 creates a small isolation impedance. Together with 10 nF C44, nominal pole 338.6 kHz, well above 50 Hz design band. This is a board-level damping/charge-delivery choice. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	CRCW / BUF / ADC2 **Static check** **Qualify on PCB**
TP6 P_DIV P_DIV	**Probe pressure divider output.** 2 mm pad on P_DIV should show about 1.246 V for a true 5 V input. It loads a24.9k source, so probe impedance matters. **Verify:** Use a high-impedance probe; include probe loading when comparing sensor/ADC values.	ADC2 **Probe geometry**
TP7 P_DRDY_N P_DRDY_N	**Probe pressure conversion timing.** 2 mm pad on DRDY exposes nominal 1 ms conversions. Clock tolerance and filter latency differ from ADS1256. **Verify:** Verify timestamp alignment, dropped samples and internal-clock drift.	ADC2 **Probe geometry**
U7 ADS1220IPWR ADS1220IPWR	**Single-ended pressure conversion.** ADS1220IPWR PW 16: AIN0=11, AVDD12,DVDD13. Gain 1/PGA bypass/internal 2.048 V/internalclock/normal 1000 SPS. CLK3,DGND4,AVSS5 are grounded; unused 6/7/8/9/10 are NC per section 9.1.5. IDAC and low-side switch OFF. **Verify:** Verify internal-reference calibration/drift, data timing and low-impedance drive from U10.	ADC2 **Static check** **Qualify on PCB**
U10 OPA388IDBVR OPA388IDBVR	**Buffers the pressure divider before the ADC.** OPA388IDBVR DBV 5: OUT1,V-2,+IN3,-IN4,V+5. At gain 1, input bias max 400pA 0-85C gives<10uV through 24.925k. Offset max 7.5uV overfulltemperature is negligible beside 1.246 VFS. This removes the direct high-impedance ADS1220 drive. **Verify:** Verify step/settling, overload recovery and stability with R46/C44; normal input stays 0-1.25 V on ~5 V supply.	BUF / ADC2 **Static check** **Qualify on PCB**

6. Radio: flow control and independent power

Native sheet: radio.kicad_sch

The owner selected XB3-24AST-J through-hole RP-SMA with A 24-HASM-450 antenna. This is the RP-SMA variant. Digi hardware manual W governs the cited power/pin recommendations; the older 8.2 pF recommendation is not silently mixed with W 47 pF.

C52=47 pF is closest to VCC; C50=1 uF and C51=10 uF handle progressively slower supply disturbance. The 1.1 MHz buck meets the switching-frequency recommendation. U3 and bypasses must still achieve≤50 mV p-p at the module under RF load/startup.

TXU0204 U9 uses VCCA from Teensy and VCCB from RF. Two channels carry MCU TX/RTS into XBee DIN/RTS; the other two carry DOUT/CTS back. It has specified partial-power-down/VCC isolation when either supply is off. C53/C54 bypass both domains. CommonGND remains.

100k pulls set TX/RX high and flow-control lines deasserted on the MCU side. When RF is off, CTS high prevents Serial3 transmission. TI recommends keeping inputs low before deliberately disconnecting a supply; verify uncontrolled rail ramps too. Reset 10k pull-up and DTR10k pull-down provide chosen boot defaults; recovery access uses TP8/TP9 plusUART/RTS.

Digi p101 requires attention to early SoC revisionB: supply dips in roughly 1.2-1.8 V can leave the module unresponsive until a complete powercycle. Specify/inspect revisionC or later, which fixes the issue. The user module label has not been checked.

For USB-adapter firmware recovery, remove socketed U8 and use the module adapter. TP8/TP9 are diagnostic access, not a complete programming header. With both supplies present U9 drives RF-side UART/RTS; external adapter signals must not contend with it.

The stated USB adapter is assumed to be the ground-station device. Its own 3.3 V output is never joined to the board regulator output. Confirm antenna/RP-SMA projection, module socket contact fit and enclosure/cable access during physical review.

Component register / 6. Radio

Values and specifications are read from native CAD. Every reference on radio.kicad_sch is listed below.

Reference / value	Purpose and value basis	Evidence / status
C50 1 uF C0603C105K4RACTU	**XBee medium-frequency supply bypass.** 1 uF 16 V X7R 0603 follows Digi revisionW p52 alongside 47 pF. Exact KEMET order/curves are archived. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / RF **Static check** **Qualify on PCB**
C51 10 uF C0805C106K4PACTU	**XBee local supply reservoir.** 10 uF 16 V X5R 0805 follows Digi additional reservoir recommendation. Effective C and actual LDO/PCB response determine RF voltage ripple. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / RF / LDO **Static check** **Qualify on PCB**
C52 47 pF VJ0603A470JXACW1BC	**XBee high-frequency VCC bypass.** 47 pF 50 V C0G 5%0603 Vishay order implements Digi 90001543 W p52. Older 8.2 pF manuals are retained as a revision conflict; the cited archivedW recommendation is followed. **Verify:** Place closest to U8 VCC/GND; measure ≤50 mV p-p at module.	VJ / RF **Static check** **Qualify on PCB**
C53, C54 (x 2) 100 nF C0603C104K5RACTU	**Independent local bypasses for UART buffer supplies.** 100 nF 50 V X7R 0603 for each supply pin 1/14, with commonGND7. TI recommends local bypass on both supplies; do not join the two 3.3 V outputs. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	KEMET / ISOLATOR **Static check** **Qualify on PCB**
R50 10 kOhm CRCW060310K0FKEA	**Holds XBee active-low RESET inactive.** 10 kOhm external pull-up is a deliberate reset default/recovery choice; not a manufacturer-required exact resistance. **Verify:** Check reset leakage/thresholds, ability of service adapter to pull low and reset timing.	CRCW / RF **Static check** **Qualify on PCB**
R51 10 kOhm CRCW060310K0FKEA	**Holds DTR/SLEEP_RQ low when no adapter drives it.** 10 kOhm pull-down establishes non-requested pin sleep state only for the selected firmware configuration. Digi requires DTR for firmware recovery. **Verify:** Configure sleep explicitly and prove external adapter high/low levels against this pull-down.	CRCW / RF **Static check** **Qualify on PCB**
R52, R53, R54, R55 (x 4) 100 kOhm CRCW0603100KFKEA	**Defines UART idle and deasserted flow-control states at MCU side.** 100k1% pulls TX/RX/RTS/CTS to Teensy3V3. TI integrated input pulldowns are 5MOhm typical; external pull-up should be≤1MOhm.100k is comfortably below that guidance and draws 33uA when low. CTS high stops the supportedSerial3 transmitter when RF is off. **Verify:** Verify boot/default levels and high-level leakage margins at temperature.	CRCW / ISOLATOR / MCU **Static check** **Qualify on PCB**
TP8 XB_RESET_N XB_RESET_N	**Expose XBee reset and DTR for service/recovery.** 2 mm test pads permit access without driving these signals from the Teensy. Probe pads are not a complete keyed programming connector. **Verify:** Verify service harness, RTS/UART access and adapter supply isolation.	RF **Probe geometry**
TP9 XB_SLEEP_RQ XB_SLEEP_RQ	**DTR / SLEEP_RQ service probe.** 2 mm pad is a mechanical/probing choice. Shares the recovery and adapter-isolation checks described for TP8. Confirm service harness and RTS/UART access.	RF **Probe geometry**
U8 XB3-24AST-J XB3-24AST-J	**Owner-selected XBee telemetry module.** XB3-24AST-J is standard through-hole RP-SMA with A 24-HASM-450. UART DIN3/DOUT2,RTS16input/CTS12output. Digi revisionW specifies 1 uF+47 pF with 10 uF reservoir; all are now present. **Verify:** Specify/inspect SoC revisionC or later per Digi p101. Earlier revisionB can latch after supply dips and require a complete powercycle. Configure DIO/sleep/flow-control; verify ripple, recovery and antenna fit.	RF **Static check** **Qualify on PCB**

Reference / value	Purpose and value basis	Evidence / status
U9 TXU0204PWR TXU0204PWR	**Buffers four UART/flow-control signals across independently powered rails.** TXU0204PWR PW 14: two channels A-to-B and two B-to-A; VCCA Teensy3V3,VCCB RF3V3,OE tiedVCCA. Datasheet specifies Hi-Z outputs when either VCC<100 mV/disconnected and Ioff power-off behavior. Common ground remains; this is not galvanic isolation. **Verify:** Check all battery/USB/off/ramp states and idle pulls. Keep inputs low before intentionally disconnecting a rail as TI recommends; verify power-down ramps.	ISOLATOR / RF **Static check** **Qualify on PCB**

7. Ignition interface: two separate current paths

Native sheet: ignition.kicad_sch

The control path is 5 V relay coil -> AO3400A -> electronicsGND. The contact path is separate 2S battery -> external physical ARM -> normally-open relay -> nichrome -> IGN_GND. There is no drawn connection between IGN_GND and electronicsGND. Contact isolation and physical clearances still require layout inspection.

R60=1k and R61=10k give nominal 3.0 V gate drive from 3.3 V. AO3400A has RDS(on) max 48 mOhm at 2.5 V, so the choice uses a specified gate condition. Pull-down defaults the gate off while the MCU is high impedance; it cannot defeat a stuck-high output.

Coil 125 Ohm +/-10% draws~46 mA at the screened upper 5 V rail. D60 cathode is+5V,anode coil_low; it recirculates turn-off current and delays relay release. Verify repeated energy and release behavior. Hongfa unlabeled type 1 drawing is converted from bottom view to PCB top view; local IDs1/6 identify coil and 2/3 identify NO contacts.

Because no load measurement is available, the author contract is≤2 A at 8.4 V, requiring cold resistance≥4.2 Ohm and a bounded pulse verified on a non-energetic load. This is an assumption about the future load, not proof that the actual nichrome meets it or heats sufficiently. Relay 5 A/30 VDC resistive rating and connector ratings do not qualify the whole harness/copper path.

J5 now follows the Kangnex 1.50 mm drill drawing. J6 is PCB-mount XT60PW-F. All three AMASS connectors usepad 2 positive/pad 1 return, matching the assigned footprint markings. AMASS revision differences, polarity, externalfuse/ARM switch, wire, strain relief and copper remain layout/procurement checks. There is no continuity or physical ARM readback and no live ignition approval.

Component register / 7. Ignition interface

Values and specifications are read from native CAD. Every reference on ignition.kicad_sch is listed below.

Reference / value	Purpose and value basis	Evidence / status
D60 1N4148WS 1N4148WS	**Relay flyback diode.** Semtech1N4148WS: cathode 1 to+5V,anode 2 tocoil_low. Initial coil current~46 mA is below 150 mA average datasheet rating; VFmax 1 V at 50 mA implies clamp near 6.17 V, below 30 V MOSFET rating. Coil energy/repetition/release delay still require testing. **Verify:** Verify PCB placement, temperature, startup and measured behavior within the stated design assumptions.	DIODE / RELAY / MOS **Static check** **Qualify on PCB**
J4 XT30PW-F XT30PW-F	**Electronics 4S / separateignition 2S battery connectors.** XT30PW-F selected drawings/5mmpowerpitch. Input J1return isGND;J4return is IGN_GND. Pad 2 is positive and pad 1 is return, matching the assigned KiCad footprint +/- markings. Connector family 15 A rating exceeds planning currents; mating gender/polarity and inconsistent retention-hole drawings require exact purchased-part fit verification. **Verify:** Do not use genericcatalog dimensions interchangeably with V 1.2; verify harness/polarity and external fuse.	XT30 **Static check** **Qualify on PCB**
J5 ARM / WJ500V-5.08-2P WJ500V-5.08-2P-14-00A	**External physical ARM switch loop.** Exact green single-pitch WJ500V-5.08-2P-14-00A. Manufacturer drawingA specifies 5.08 mm pitch and 1.50 mm finishedholes. Imported pads already had 5.08 pitch despite 5.00 name, but 1.30 mm drills differed. New local footprint uses 1.50 mm drills/3 mm pads. UL10 A/300 V contact rating exceeds assumed 2 A/8.4 V. **Verify:** Verify purchased drawingrevision, screw access/0.4Nm torque/6-7 mm strip, actual fit and series ARM switch.	ARM **Static check** **Qualify on PCB**
J6 XT60PW-F XT60PW-F	**PCB-mount ignition output in the chosenXT60 family.** XT60PW-F replaces wire-termination XT60-F with a defined horizontal PCB part. V 1.2 installation drawing matches 7.20 mm powerpitch/2.70 mm holes and 13.5 mm retention-slot separation/6 mm setback. Pad 2 is switched positive, pad 1 is IGN_GND, matching footprint +/- markings. Newer catalog body dimensions differ; order/fit must be verified. **Verify:** Use conservative 30 A catalog rating only as connector evidence; systempath remains 2 A design assumption, subject to fuse/switch/copper validation.	XT60 / RELAY **Static check** **Qualify on PCB**
K1 HF46F-G/5-HS1T HF46F-G/5-HS1T	**Normally-open relay contact in the separate ignition path.** HF46F-G/5-HS1T: local pad IDs1/6 for coil and 2/3 for NO contacts; Hongfa type 1 bottom-view drawing verifies topology/coordinates but does not number the terminals;5 V/125 Ohm +/-10%. At5.1704 V and 112.5 Ohm, coil current 45.96 mA fits 47 mA plan. Manufacturer 5 A resistive/30 VDC exceeds assumed 2 A/8.4 V; actual load is unknown. **Verify:** Qualify real current, duration, temperature, contact release/faults and physical footprint with a dummy load; nolive-load acceptance.	PART_HF46F_G_5_HS1T / RELAY **Static check** **Qualify on PCB**
Q1 AO3400A AO3400A	**Low-side relay-coil switch.** AO3400A is specified up to 48 mOhm RDS(on) at VGS=2.5 V, below the nominal 3.0 V drive. VGS(th) alone is not used to claim full turn-on. **Verify:** Confirm exact manufacturer/package, pulse/current/thermal and reset/fault behavior; it does not switch the ignition load directly.	MOS **Static check** **Qualify on PCB**
R60 1 kOhm CRCW06031K00FKEA	**Limits MOSFET gate charge current/edge speed.** 1 kOhm plus 10 kOhm pull-down forms a DC divider: VGS approximately 3.3*10/11=3.0 V. Value is an engineering choice, not an AO3400A requirement. **Verify:** Check MCU VOH, transition time, gate leakage and brownout; GPIO supplies ~0.30 mA steady when high.	CRCW / MOS / MCU **Static check** **Qualify on PCB**
R61 10 kOhm CRCW060310K0FKEA	**Keeps the MOSFET off with a floating MCU output.** 10 kOhm discharges the gate and overcomes leakage. Together with R60 it reduces the driven gate voltage to about 3.0 V. **Verify:** Verify actual GPIO reset states and noise immunity; a resistor cannot defeat a stuck-high output.	CRCW / MOS **Static check** **Qualify on PCB**
TP10 COIL_LOW COIL_LOW	**Probe relay drain/coil-low signal.** 2 mm pad helps verify commanded ON/OFF, flyback and reset behavior with a dummy load. It is not a current measurement. **Verify:** Use safe probing relative to electronics GND; confirm no accidental bridge to IGN_GND.	MOS / DIODE / RELAY **Probe geometry**

Support-component completeness

This is the required-versus-present cross-check. Optional and module-owned functions are distinguished from omissions. Pin inventory and ERC are separate checks.

Device / required function	Current implementation	Result / datasheet basis
LMR51430 power stage	C1/C11/C12 input; C2 HF; C3 CB-SW; L1; ten output MLCCs; R1/R2 and R3/R4/R5	**Present; equation screen conditional** BUCK pp 16-20. C9 removed for the new topology. D2 is an author-added bounded clamp.
TLV75533 stability	C7/C8 exact 10 uF 16 V; IN1/EN3 to 5 V; GND2; OUT5; NC4	**Present** LDO pp 3,15-16; effective C/ripple measured on prototype.
Teensy module support	VIN reservoir/shunt;48 signal-named socket pads; module owns MCU regulator/clock/SD socket	**Present; USB configuration required** PJRC cards/power/UART pages; do not duplicate module support.
OPA2388 bridge buffers	100 nF supply; two unity feedback loops;1k/220 nF input filter;47 Ohm/10 nF ADC drive	**Present** BUF/ADC1; normal-input and overload qualification are separate.
ADS1256 supplies/control	AVDD1 C23/C27; DVDD16 C24/C28; GND2/17; RESET15/SYNC14 high	**Present** ADC1 pin table/basic circuit. Digital IO states configured by firmware.
ADS1256 reference/clock	Real buffered VEXC/2 reference;47 uF/100 uF/100 nF bank;7.68 MHz crystal/30 pF loads	**Present; stability/startup conditional** ADC1 Fig 25/26, REF/XTAL. No internal precision reference is assumed.
ADS1256 unused pins	AINCOM5/AIN2-7 pins 8-13 and DIO25-28 explicitly NC	**Present** Only AIN0/AIN1 selected.
OPA350 reference buffer	Unity feedback;100 nF supply;15ms input filter; exactPanasonicFT output bank	**Present; loop qualification open** REF and ADC1Fig26; ESR maxima known, minima/loop response unproven.
Pressure sensor interface	9-32 V sensor supplied from protected 4S; board-to-DIN1/3/2 harness;100k/33.2k/47 nF	**Present; sensor Rs assumption** SENSOR pp 1-3; 133.2k load not specified by sensor datasheet.
OPA388 pressure buffer	Unity feedback;C43 bypass;R46/C44 local ADC drive	**Present** BUF DBV 5 table/layout and ADC2 input-current discussion.
ADS1220 supplies/input	AVDD12 C41;DVDD13 C42;CLK3/DGND4/AVSS5 GND;AIN0 pin 11	**Present** ADC2 PW 16 table. PGA bypass required for single-ended-to-AVSS.
ADS1220 unused pins	6/7/8/9/10 NC;low-side switch/IDAC OFF;all digital outputs accounted for	**Present** ADC2 section 9.1.5.
XBee supply/recovery	47 pF+1 uF+10 uF;RESET pull-up;DTR pull-down;testpads/UART/RTS access	**Present; recovery/fit conditional** Digi 90001543 W pp 45-52.
TXU0204 supplies/enable	C53/C54 separate 100 nF;VCCA 1/OE8 Teensy3V3;VCCB14 RF3V3;GND7;NC6/9	**Present** ISOLATOR PW 14 table and power-off conditions.
Relay switch/flyback	Gate resistor/pull-down;MOSFET;flyback diode;physical externalARM and separateIGN_GND	**Present; system qualification open** RELAY/MOS/DIODE. No continuity/ARM sensing requested.
Connector footprints	All 120 pin sets match actual pads; shared geometry parses all	**Source-bound pad coverage passes; fit open** Five localTMS footprints have validated manufacturer-drawing-derived receipts; these are author lineage records, not physical-fit acceptance.

External and assembly parts

The 120 CAD references are not the complete procurement list. A module footprint represents one schematic component; female socket rows, inserted SD card, antenna, cables and external supplies are separate items. Quantities marked TBD cannot be safely invented.

Item / quantity	Purpose	Remaining verification
BCA-100L 1	Confirmed external load cell; not a board-mounted component	Verify actual model/capacity, conductors, calibration and 5 V excitation performance.
IPSU-GP1K5-6 1	Confirmed external pressure sensor	Verify actual unit, DIN mating cable, supply/return/signal and loading.
A 24-HASM-450 1	Confirmed external antenna	Verify RP-SMA mate, mounting and enclosure clearance; extension optional.
microSD card 1	Inserted into U1 module SD socket	Select exact card; verify write-latency/peak-current/endurance and sustained logging.
Teensy female socket rows 2 x 24 contacts	Physical carrier sockets represented by U1 footprint	Select exact 2.54 mm sockets, height, retention and current/contact ratings.
XBee female socket rows 2 x 10 contacts	Physical sockets represented by U8 footprint	Select exact 2 mm sockets and correct 22 mm row separation/height.
JP1 shunt 1	Removable VIN link	Select exact mating shunt/current rating and define USB procedure.
Battery packs 1 x 4S and 1 x 2S	Separate electronics and ignition supplies	Select packs/protection from actual loads; capacities and peak ratings unresolved.
External fuses/holders TBD per supply/harness	Not present on PCB	Select from measured current, wire ratings, battery fault current and clearing characteristics.
Physical ARM switch 1	External series switch throughJ5	Current/voltage/pulse and environmental ratings unresolved.
Nichrome/load assembly 1 test assembly	External ignition load	Cold/hot resistance, construction, pulse energy and duration unresolved.
Harness/mating connectors 1 mate per J1-J6, where applicable	XT30 x 2, XT60 x 1, XH5 x 1, XH3 x 1; J5 switch wiring	Verify actual connector gender, crimp contacts, wire size/length, strain relief and shield termination.
XBee USB adapter 1 ground-station unit assumed	Owner stated 3.3 V supply	Exact adapter/current capacity/install location unresolved; do not parallel supply outputs.
Enclosure/mounting/antenna extension TBD; extension 0 or 1	Not established by schematic	Resolve 100x80 versus 100x100 mm envelope and mounting conflict before PCB placement.

Pin-by-pin device inventory

These active-device/module and relay pin mappings are read directly from the saved native netlist. Compare the pin numbers with the selected datasheet. Generic auto-net names are preserved to avoid pretending they were engineering labels. The CSV contains every passive/connector/test-point pin as well.

U1 / TEENSY41

Pin	Native net / NC	Pin	Native net / NC
3V3L	+3V3	D22	NC
3V3R	+3V3	D23	NC
D0	P_CS_N	D24	NC
D1	P_MISO	D25	NC
D2	P_DRDY_N	D26	P_MOSI
D3	FIRE_CMD	D27	P_SCK
D4	Net-(U1-D4_{slash}_STATUS)	D28	NC
D5	Net-(U1-D5_{slash}_COMMAND)	D29	NC
D6	Net-(U1-D6_{slash}_TX_LED)	D30	NC
D7	NC	D31	NC
D8	NC	D32	NC
D9	LC_DRDY_N	D33	NC
D10	LC_CS_N	D34	NC
D11	LC_MOSI	D35	NC
D12	LC_MISO	D36	NC
D13	LC_SCK	D37	NC
D14	XB_TX	D38	NC
D15	XB_RX	D39	NC
D16	NC	D40	NC
D17	NC	D41	NC
D18	XB_RTS_N	GNDA	GND
D19	XB_CTS_N	GNDL	GND
D20	NC	GNDR	GND
D21	NC	VIN	Teensy controller/TEENSY_VIN

U2 / LMR51430YFDDCR

Pin	Native net / NC	Pin	Native net / NC
1	GND	4	Net-(U2-FB)
2	Net-(U2-SW)	5	Net-(U2-EN)
3	VBAT	6	Net-(U2-CB)

U3 / TLV75533PDBVR

Pin	Native net / NC	Pin	Native net / NC
1	+5V	4	NC
2	GND	5	+3V3_RF
3	+5V		

U4 / OPA2388IDR

Pin	Native net / NC	Pin	Native net / NC
1	Net-(U4-OUT_A)	5	Net-(C20-Pad2)
2	Net-(U4-OUT_A)	6	Net-(U4-OUT_B)
3	Net-(C20-Pad1)	7	Net-(U4-OUT_B)
4	GND	8	+5VA

U5 / ADS1256IDBR

Pin	Native net / NC	Pin	Native net / NC
1	+5VA	15	+3V3
2	GND	16	+3V3
3	GND	17	GND
4	Ratiometric reference/LC_VREF	18	Net-(U5-XTAL2)
5	NC	19	Net-(U5-XTAL1_{slash}_CLKIN)
6	Net-(U5-AIN0)	20	LC_CS_N
7	Net-(U5-AIN1)	21	LC_DRDY_N
8	NC	22	Net-(U5-DOUT)
9	NC	23	Net-(U5-DIN)
10	NC	24	Net-(U5-SCLK)
11	NC	25	NC
12	NC	26	NC
13	NC	27	NC
14	+3V3	28	NC

U6 / OPA350UA

Pin	Native net / NC	Pin	Native net / NC
1	NC	5	NC
2	Ratiometric reference/LC_VREF	6	Ratiometric reference/LC_VREF
3	Net-(U6-+)	7	+5VA
4	GND	8	NC

U7 / ADS1220IPWR

Pin	Native net / NC	Pin	Native net / NC
1	Net-(U7-SCLK)	9	NC
2	P_CS_N	10	NC
3	GND	11	Net-(U7-AIN0_{slash}_REFP1)
4	GND	12	+5VA
5	GND	13	+3V3
6	NC	14	P_DRDY_N
7	NC	15	Net-(U7-DOUT_{slash}_~{DRDY})
8	NC	16	Net-(U7-DIN)

U8 / XB3-24AST-J

Pin	Native net / NC	Pin	Native net / NC
1	+3V3_RF	11	NC
2	XBee telemetry/RF_DOUT	12	XBee telemetry/RF_CTS_N
3	XBee telemetry/RF_DIN	13	NC
4	NC	14	NC
5	Net-(U8-~{RESET})	15	NC
6	NC	16	XBee telemetry/RF_RTS_N
7	NC	17	NC
8	NC	18	NC
9	Net-(U8-DTR_{slash}_SLEEP_RQ)	19	NC
10	GND	20	NC

U9 / TXU0204PWR

Pin	Native net / NC	Pin	Native net / NC
1	+3V3	8	+3V3
2	XB_TX	9	NC
3	XB_RTS_N	10	XBee telemetry/RF_CTS_N
4	XB_RX	11	XBee telemetry/RF_DOUT
5	XB_CTS_N	12	XBee telemetry/RF_RTS_N
6	NC	13	XBee telemetry/RF_DIN
7	GND	14	+3V3_RF

U10 / OPA388IDBVR

Pin	Native net / NC	Pin	Native net / NC
1	Net-(U10--)	4	Net-(U10--)
2	GND	5	+5VA
3	Net-(U10-+)		

Q1 / AO3400A

Pin	Native net / NC	Pin	Native net / NC
1	Net-(Q1-G)	3	Net-(D60-A)
2	GND		

K1 / HF46F-G/5-HS1T

Pin	Native net / NC	Pin	Native net / NC
1	+5V	3	Net-(J6-+)
2	Net-(J5-ARM_RETURN)	6	Net-(D60-A)

Source index

Supplemental primary source: PJRC hardware serial table, https://www.pjrc.com/teensy/td_uart.html, retrieved 14-Oct-2026. Exact HTML is retained in docs/research/pjrc-uart-20261014.html. Serial3 directCTS19 is documented there.

Page locations refer to the archived PDF, not mutable online pagination. Exact archived hashes are recorded in evidence/index.yaml and the accompanying Markdown. Click source labels to open the publisher/supplier page; use the local archived copy when comparing revision-specific values.

BUCK / lmr 51430.pdf

SLUSEF4 A: pp 3-6 pin map/limits; pp 16-20, Table 9-2, equations 11-17; DDC lands

Publisher / supplier source
evidence/files/LMR51430YFDDCR_lmr51430_f7d8053844fb.pdf

LDO / tlv 755p.pdf

pp. 3-6, 15-16, 19; DBV package

Publisher / supplier source
evidence/files/TLV75533PDBVR_tlv755p_d5d9f30657f2.pdf

ADC1 / ADS1256_ti

pp. 3-5, 14-18, 27-29; Fig. 25-26

Publisher / supplier source
evidence/files/ADS1256_ti.pdf

ADC2 / ADS1220_ti

pp. 3-7, 20-26, 35-42, 49-50, 64-65

Publisher / supplier source
evidence/files/ADS1220_ti.pdf

BUF / opa 2388.pdf

pp. 3, 7-9, 15, 24-26

Publisher / supplier source
evidence/files/OPA2388IDR_opa2388_bfaa58483429.pdf

REF / opa 350.pdf

pp. 3, 6, 12, 15-17, 22-23

Publisher / supplier source
evidence/files/OPA350UA_opa350_185293ca6674.pdf

RF / xbee 3_hardware_ref

90001543 W (June 2026), pp. 26-27, 45-47, 51-52

Publisher / supplier source
evidence/files/xbee3_hardware_ref.pdf

MCU / card 11a_rev4_web

pp. 1-2; companion rear card and PJRC power page

Publisher / supplier source
evidence/files/card11a_rev4_web.pdf

BRIDGE / BCA.pdf

p. 2 specifications and wiring

Publisher / supplier source
evidence/files/BCA-100L_BCA_b7855a5002cd.pdf

SENSOR / 4da886fe05e2bf87f204aeba6d317782.pdf

pp. 1-2 family/order/wiring

Publisher / supplier source
evidence/files/IPSU-GP1K5-6_4da886fe05e2bf87f204aeba6d317782_5914137a3440.pdf

XTAL / ABLS.pdf

pp. 1-3; CL/drive/order options

Publisher / supplier source
evidence/files/ABLS-7.680MHZ-D-4-Y-T_ABLS_649cda5a6e08.pdf

IND / srp 1038a.pdf

pp 1-2; exact -3R3M row: 3.3uH +/-20%, 11.8 mOhm max DCR, Irms 11 A/Isat 20 A at 25C

Publisher / supplier source
evidence/files/SRP1038A-100M_srp1038a_92c761c358a1.pdf

ESD / TPD2E2U06 TI datasheet (DCK 3-pin package selected)

pp. 3-4, 10-12; actual DCK 3-pin variant

Publisher / supplier source
evidence/files/TPD2E2U06DBZR_tpd2e2u06_a133b86ea3d3.pdf

CLAMP / BAV199.pdf

pp. 1-3; pin table and leakage/VF limits

Publisher / supplier source
evidence/files/BAV199_BAV199_ab2f26d003e3.pdf

MOS / AO3400A.pdf

pp. 1-3; RDS(on) at VGS=2.5 V

Publisher / supplier source
evidence/files/AO3400A_AO3400A_9c60d0b6c1dd.pdf

RELAY / HF46F_hongfa_en

pp. 1-2; 5 V coil and type-1 bottom view

Publisher / supplier source
evidence/files/HF46F_hongfa_en.pdf

DIODE / 1N4148WS_C57759

pp. 1-3; Semtech Electronics SC-76

Publisher / supplier source
evidence/files/1N4148WS_C57759.pdf

REVERSE / SS54_C22452

pp. 1-2; MDD SS54 SMA

Publisher / supplier source
evidence/files/SS54_C22452.pdf

GREEN / APTD1608LCGCK.pdf

pp. 1-2; low-current green

Publisher / supplier source
evidence/files/APTD1608LCGCK_APTD1608LCGCK_7bb244fd771d.pdf

RED / APTD1608LSURCK.pdf

pp. 1-2; low-current red

Publisher / supplier source
evidence/files/APTD1608LSURCK_APTD1608LSURCK_8c4ae4d0cb2c.pdf

YELLOW / APTD1608LSYCK.pdf

pp. 1-2; low-current yellow

Publisher / supplier source
evidence/files/APTD1608LSYCK_APTD1608LSYCK_9fe32571563b.pdf

JST / eXH.pdf

pp. 1-3; XH family and B3B/B5B drawings

Publisher / supplier source
evidence/files/B5B-XH-A_eXH_9426b136902f.pdf

XT30 / XT30PW-F.pdf

p. 1; exact connector drawing

Publisher / supplier source
evidence/files/XT30PW-F_XT30PW-F_090aabb47c14.pdf

XT60 / XT60PW-F.pdf

AMASS V 1.2 p1 installation-hole drawing; also retain LCSC two-page specification with dimensional differences

Publisher / supplier source
evidence/files/XT60PW-F_XT60PW-F_5d9495ef6e83.pdf

ARM / 5a3d038c44f6e3872ef57456c5bdf975.pdf

Drawing A 2024-03-10, p1: full ordering code; 5.08 pitch; 1.50 mm holes; UL10 A; body 10.16x10x14.07 mm

Publisher / supplier source
evidence/files/WJ500V-5.08-2P-14-00A_5a3d038c44f6e3872ef57456c5bdf975_e3c3cbd8e0f7.pdf

ISOLATOR / txu 0204.pdf

SCES936 A pp 4-9, 21-24: PW 14 pins, Ioff/VCC isolation, integrated pulldowns, supply bypasses

Publisher / supplier source
evidence/files/TXU0204PWR_txu0204_d54f3853ad68.pdf

TVS / SMBJ5.0(C)A-SMBJ170(C)A.pdf

DS19002 Rev 20-2 pp 1-3,5-6: unidirectional ordering, pulse conditions, SMBJ18 A electrical row and SMB lands

Publisher / supplier source
evidence/files/SMBJ18A-13-F_SMBJ5.0_C_A-SMBJ170_C_A_752945fa3aca.pdf

CRCW / dcrcwe 3.pdf

Vishay 20035 14-Apr-2026 pp 1-3: 0603, F=1%, K=100 ppm/K, 0.125 W at 70C, 75 V; order table

Publisher / supplier source
evidence/files/CRCW06031K00FKEA_dcrcwe3_dd2d82bab2e2.pdf

TNPW / tnpw_e3.pdf

Vishay 28758 10-Apr-2026 pp 1-4: B=0.1%, E=25 ppm/K, 0603; 0.210 W at 70C/100 V subject to thermal management

Publisher / supplier source
evidence/files/TNPW0603100KBEEA_tnpw_e3_6f0508c568cb.pdf

TNPU / tnpue 3.pdf

Vishay 28779 04-Mar-2025 pp 1-3: H=0.02%, W=2 ppm/K, 10K0; 0603 0.1 W at 70C; two discrete resistors are not a matched network

Publisher / supplier source
evidence/files/TNPU060310K0HWEA00_tnpue3_d1f1bfcd0370.pdf

VJ / vjw 1bcbascomseries.pdf

Vishay 28548 12-Aug-2026 pp 2-6: C0G, J=5%, X=50 V, 0603 size/capacitance matrix and lands

Publisher / supplier source
evidence/files/VJ0603A470JXACW1BC_vjw1bcbascomseries_757d89d0506d.pdf

FT / ABA 0000C1240.pdf

Panasonic FT ABA 0000C1240, p3: 47 uF 16 V 4x5.8 mm and 100 uF 16 V 5x5.8 mm; ESR max 0.85/0.36 Ohm at 100 kHz 20C; ripple 160/240 mA; +/-20%,105C/2000h

Publisher / supplier source
evidence/files/EEEFT1C101AR_ABA0000C1240_affbdbc3402d.pdf

FERRITE / Murata-BLM21PG601SN1-Series-EN-Datasheet.pdf

Murata manufacturer ProductSearchDataSheet mirrored by Avnet pp 1-4; 600 Ohm at 100 MHz; compare current JENF243 A_0005 AE-01

Publisher / supplier source
evidence/files/BLM21PG601SN1D_Murata-BLM21PG601SN1-Series-EN-Datasheet_ffa5cd2dd18c.pdf

SHUNT_HEADER / tsw_th.pdf

Samtec TSW through-hole family drawing; 2 contacts/single row/2.54 mm; -07 tail choice and L plating

Publisher / supplier source
evidence/files/TSW-102-07-L-S_tsw_th_3a5770b11668.pdf

C0603C104K5RACTU / C0603C104K5RACTU

Exact KEMET part specification generated 14-Oct-2026: p1 guaranteed size/value/rating/tolerance; pp 2-3 typical simulation curves, not limits; p4 land pattern

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C0603C105K4RACTU / C0603C105K4RACTU

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C0603C224K4RACTU / C0603C224K4RACTU

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C0603C300J5GACTU / C0603C300J5GACTU

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C0805C106K4PACTU / C0805C106K4PACTU

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C1206C473J5GACTU / C1206C473J5GACTU

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C1210C106K5RACTU / C1210C106K5RACTU

Exact KEMET part specification generated 14-Oct-2026: p1 guaranteed size/value/rating/tolerance; pp 2-3 typical simulation curves, not limits; p4 land pattern

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PART_HF46F_G_5_HS1T / HF46F_C5274026

Manufacturer datasheet: selected order, package/pin map and electrical characteristics

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evidence/files/HF46F_C5274026.pdf